

LEVEL 7 · WRITE-UP

First Contact

A seven-stage finale: galactic-dynamics chain -> invert a modular cipher (RE + maths) -> Arecibo bitmap -> OSINT (Vega) -> MOND derivation -> XOR-unlock a Pastebin -> known-plaintext attack -> flag. No stage is a black box and none can be brute-forced.

Stage 1 — the physics chain

```
v_r = -c*df/f0 = -(3.00e5)(-521e3)/1.420e9 = 110 km/s -> 110
v_c = v_r + v_sun*sin(l) = 110 + 220*0.5 = 220 km/s -> 220
R = R_sun*sin(l) = 4.00 kpc; M = v_c^2 R/G = 45 (1e9 Msun) -> 45
T = 2*pi*R/v_c = 112 Myr -> 112
KEY = 110 220 45 112 = 11022045112
```

Stage 2 — inverting the machine (RE + modular inverse)

generator.py maps the message to blocks of a big integer and multiplies each by $A = (\text{KEY}^3 \text{ XOR } 0x9E37...) \bmod P$, with $P = 2^{521} - 1$. It runs forward only; running it again multiplies by A a second time. To decrypt you must multiply by the modular inverse $A^{-1} \pmod{P}$ — a value you compute, not one the program gives you. That is the point: no plug-and-run, no brute force.

```
A = (pow(KEY,3,P) ^ 0x9E3779B97F4A7C15) % P # read from generator.py
Ainv = pow(A, -1, P) # modular inverse
blocks_clear = [(e * Ainv) % P for e in blocks] # -> 1679 bits
```

Stage 3 — the Arecibo bitmap

1679 = 23×73 (both prime) -> arrange 23 wide (73 is noise). The picture: digits 0-9, DNA, a human, an eye, a knife, then coordinate rows and a name row:

```
lat -> 10010 = 18
lon -> 1000010 = 66 => 18N 66W = Arecibo Observatory
name -> 10110 00101 00111 00001 = 22,5,7,1 = V,E,G,A -> VEGA
```

Stage 4 — OSINT

The name points to Vega (α Lyrae). Recognising it is the pivot; it is the star the finale turns on.

Stage 5 — MOND (on paper) and the waypoint

Deep-MOND, $a \ll a_0$: $\mu(x) \approx x$, so $F = m a^2/a_0$. Equating to gravity GMm/r^2 with $a = v^2/r$ gives $v^4 = G M a_0$, i.e. $v = (G M a_0)^{1/4}$ — flat, radius-independent (this is why Tully-Fisher has exponent 4). With $M = 45 \times 10^9 M_\odot$ (from Stage 1) and $a_0 = 1.2 \times 10^{-10}$:

```
v = (G*M*a0)^(1/4) = (6.674e-11 * 8.955e40 * 1.2e-10)^0.25
= 1.636e5 m/s = 164 km/s -> 164
XOR the waypoint with repeating '164':
59424041...76950 ^ 164164... -> https://pastebin.com/7dh9PsXf
```

Stage 6 — known-plaintext attack

The paste holds a hex ciphertext, XOR-encrypted with a short repeating word that is not given anywhere — it must be recovered by attack. Every flag starts CTC{; XOR that 4-byte crib against the first ciphertext bytes to reveal the first key characters — ELLI. Extend by crib-dragging (the readable plaintext guides each next character) to the full key ELLIE_ARROWWAY — the heroine of Contact, who receives the signal from the stars. Decrypt the rest:

```
ct = bytes.fromhex(open('paste').read())
key = 'ELLIE_ARROWWAY' # recovered by crib-drag from CTC{, not given
flag = ''.join(chr(ct[i]^ord(key[i%len(key)])) for i in range(len(ct)))
```

CTC{For_small_creatures_such_as_we_the_vastness_is_bearable_only_throu
gh_love}

Carl Sagan, Contact.

Builder checklist

- Ship transmission.bin + generator.py. Story in prompt.txt (paste into CTFd). Keep this write-up and _full_solve.py private.
- Pastebin (7dh9PsXf): put the flag's XOR ciphertext (hex) + the 'you know how it begins' hint. The KPA key (ELLIE_ARROWWAY) is NOT in the paste and NOT derivable from the other steps — it is recovered only by crib-dragging from CTC{.
- The MOND waypoint ciphertext and problem are in the challenge PDF; its answer (164) decodes the pastebin URL.
- Every number is pinned/deterministic: 110,220,45,112 -> KEY; MOND -> 164 (unlocks the URL only); KPA key recovered by crib.
- Flag exact-match, case-sensitive. Run _full_solve.py on a fresh machine before the event.